LAB ASSIGNMENT – 3

Course: Fundamentals of Operating System Lab (ENCA252)

Program: BCA (AI & DS)

Title: Comprehensive Implementation of Page Replacement Algorithms

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Aim

To implement and analyze various page replacement algorithms including FIFO, LRU, Optimal, MRU, and Second Chance, and compare their performance based on page faults.

Objective

- To understand virtual memory and paging

- To simulate different page replacement techniques

- To calculate page faults

- To compare efficiency of algorithms

Tools Used

- Python 3.x

- Ubuntu/Linux

- VS Code

Theory

In virtual memory systems, when all frames are occupied and a new page is required, a page replacement algorithm decides which page to remove. Different algorithms use different strategies to minimize page faults and improve system performance.

Algorithms Implemented

1. FIFO (First In First Out)

- Replaces the oldest page in memory

- Simple but may lead to higher faults

2. LRU (Least Recently Used)

- Replaces the least recently used page

- Uses past usage information

3. Optimal

- Replaces the page that will not be used for the longest time in future -

Gives minimum page faults

4. MRU (Most Recently Used)

- Replaces the most recently used page
- Opposite of LRU

5. Second Chance (Clock Algorithm)
- Modified FIFO using reference bits
- Gives second chance to recently used pages

Input
- Page Reference String:
  7, 0, 1, 2, 0, 3, 0, 4, 2, 3, 0, 3, 2
- Number of Frames: 3

Output

Algorithm	Page Faults
FIFO	10
LRU	9
Optimal	7
MRU	11
Second Chance	9–10 (approx.)

```
[Greaty7027@clouddevops:~$ nano  page_replacement.py
[Greaty7027@clouddevops:~$ python3  page_replacement.py
 Page Reference String: [7, 0, 1, 2, 0, 3, 0, 4, 2, 3, .0, 3, 2]
 Number of Frames: 3

 FIFO Faults: 10
 LRU Faults: 9
 Optimal Faults: 7
 MRU Faults: 11
 Second Chance Faults: 9
```

Analysis

- Optimal algorithm gives minimum faults because it uses future knowledge

- LRU performs better than FIFO due to locality of reference

- MRU performs worst in this case

- Second Chance improves FIFO performance

Result

The program successfully implemented all page replacement algorithms and compared their performance. Optimal algorithm showed the best performance with minimum page faults.

Conclusion

Page replacement algorithms play a crucial role in memory management. Choosing the right algorithm improves system efficiency. LRU and Optimal are generally better than FIFO.
